

MALABO, Equatorial Guinea

WMO: 648100

Lat: 3.755N

Lon: 8.709E

Elev: 76

StdP: 14.66

Time Zone: 1.00 (E01)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	71.2	71.5	67.6	101.7	75.4	68.2	104.1	75.1	16.9	80.5	16.0	80.5	3.2	230	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.2	90.0	79.9	89.6	79.9	88.8	79.8	82.8	86.4	82.1	86.0	81.4	85.8	9.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.3	168.5	85.3	80.9	161.1	84.4	80.6	159.1	84.2	46.9	86.8	46.0	87.6	45.2	86.2	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	13.4	12.0	68.9	91.7	1.0	0.6	68.2	92.1	67.6	92.5	67.1	92.8	66.4	93.3
			66.6	85.0	1.6	2.0	65.5	86.4	64.5	87.6	63.6	88.8	62.4	90.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.9	81.0	82.1	81.9	81.2	80.8	79.3	78.2	77.7	78.0	78.8	79.7	80.5	(6)
	DBStd	2.06	1.50	1.66	1.70	1.67	1.49	1.45	1.40	1.15	1.34	1.30	1.39	1.23	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	10918	960	898	990	935	955	878	873	860	841	892	891	945	(10)
	CDD65	5443	495	478	525	485	490	428	408	395	391	427	441	480	(11)
	CDH74	48540	5110	5225	5682	4997	4717	3409	2697	2384	2517	3175	3881	4746	(12)
	CDH80	14215	1876	1947	2063	1724	1489	752	366	267	338	635	1087	1671	(13)
(14) Wind	WSAvg	5.4	5.4	6.3	5.5	5.0	4.5	4.8	5.3	6.5	5.9	5.1	5.1	4.9	(14)
(15) Precipitation	PrecAvg	99.9	1.8	2.9	6.3	8.4	10.9	11.3	12.0	11.5	12.8	13.1	6.9	2.0	(15)
	PrecMax	119.6	5.5	10.1	10.0	12.7	15.5	19.7	18.2	18.5	21.9	18.8	15.2	4.7	(16)
	PrecMin	73.1	0.2	0.2	1.8	4.4	7.2	7.1	3.2	2.1	7.6	9.6	2.0	0.2	(17)
	PrecStd	9.3	1.2	2.0	2.0	1.9	2.2	2.7	4.1	3.9	3.1	1.9	2.8	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	91.4	91.3	90.6	89.9	88.2	86.2	84.6	85.7	86.4	88.2	89.8	(19)
		MCWB	79.4	79.9	79.9	80.2	79.9	80.4	78.7	77.7	78.2	79.0	80.3	79.0	(20)
	2%	DB	89.4	89.9	89.9	89.6	88.4	86.4	84.5	83.9	84.3	85.8	87.7	88.2	(21)
		MCWB	79.3	79.9	80.1	80.2	80.1	79.3	77.9	77.4	77.8	78.9	79.9	79.0	(22)
	5%	DB	88.1	89.5	89.4	88.1	87.8	85.7	83.1	82.6	82.9	84.5	86.3	87.9	(23)
		MCWB	79.0	79.9	80.1	80.1	80.1	79.0	77.2	76.8	77.4	78.4	79.4	78.9	(24)
	10%	DB	87.7	88.1	88.0	87.6	86.4	84.2	82.3	81.7	82.3	83.9	85.7	87.0	(25)
		MCWB	78.9	79.6	80.1	80.1	79.8	78.4	76.8	76.5	77.2	78.2	79.2	78.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.3	83.6	83.5	83.4	83.5	82.5	80.4	80.6	80.7	82.2	83.3	82.7	(27)
		MCDB	86.5	87.9	88.0	87.0	86.6	85.4	83.2	82.1	82.3	84.0	85.5	86.0	(28)
	2%	WB	81.0	82.0	82.3	82.3	82.0	80.9	79.2	78.9	79.2	80.5	81.5	81.0	(29)
		MCDB	86.4	86.8	86.7	86.3	85.7	84.1	82.3	81.3	81.8	83.0	84.8	86.3	(30)
	5%	WB	80.2	81.0	81.3	81.4	81.1	79.9	78.5	78.0	78.7	79.3	80.3	80.2	(31)
		MCDB	86.1	86.7	86.5	86.1	85.4	83.4	81.7	80.8	81.5	82.7	84.7	86.0	(32)
	10%	WB	79.7	80.2	80.4	80.6	80.4	79.0	77.6	77.2	77.9	78.7	79.5	79.6	(33)
		MCDB	85.5	86.0	86.1	85.9	85.0	82.6	80.9	80.3	80.9	82.5	83.9	85.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	13.2	12.6	12.7	12.2	10.6	8.9	8.6	9.0	10.4	12.0	14.0	(35)
		MCDBR	14.9	14.2	13.9	13.9	13.9	12.9	10.5	10.1	10.3	11.8	13.5	15.1	(36)
	5% WB	MCWBR	8.1	7.5	7.1	7.7	7.7	7.6	6.2	5.8	6.1	7.3	8.0	8.4	(37)
		MCWBR	14.0	13.4	12.9	13.2	12.6	11.8	9.9	9.4	9.8	10.9	12.5	14.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.804	0.820	0.788	0.559	0.500	0.532	0.537	0.533	0.455	0.460	0.577	0.734	(40)
		taud	1.545	1.527	1.585	2.063	2.248	2.138	2.122	2.133	2.379	2.383	2.002	1.656	(41)
	Ebn at Noon		189	191	199	245	253	241	242	248	273	273	240	201	(42)
		Edh at Noon	91	95	89	54	44	48	49	50	40	40	57	81	(43)
(44) All-Sky Solar Radiation	RadAvg		1539	1467	1401	1425	1321	1061	961	962	1054	1203	1321	1470	(44)
	RadStd		106	93	93	52	81	101	85	69	77	56	63	66	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page